

ARC / CIRCLE 7-WEEK HACKATHON · AGENTIC ECONOMY TRACK · FINAL SUBMISSION

ACR – the Arc Compute Rate

Machine commerce just got its **SOFR** – and it prints its own **attack cost**.

A manipulation-resistant reference rate for machine services, live on Arc testnet – and the cash-settled futures venue that trades on it.

arc-compute-rate.vercel.app · github.com/kaustubh76/ACR---Arc-Compute-Rate

THE PROBLEM

Agents trade machine services with **no trustworthy price**

- Agents already pay per call for inference, GPU time, and data – but there is **no benchmark** to settle contracts against or hedge with.
- The raw payment exhaust is **noisy, batched, and adversarial**.
- A naive volume-weighted average is an open invitation: our own red team moves it **+122%** with wash trades.

No benchmark means no hedging, no term markets, no credit. Machine commerce is stuck at spot.

Meet the hedger: an agent business with a floating USDC burn

- **Portia**, an autonomous paralegal fleet, resells LLM work at fixed prices but pays for inference per call – **~100M tokens/mo × 0.49236 \$/1k tokens** (the ACR-INF print) **≈ \$49,236/mo**, all floating. One +10% month is **+\$4,924** of unbudgeted burn.
- The fix: go **long ACR-INF futures**. Cash-settled – at expiry the venue **freezes the freshest oracle print** (≤ 2 h old, enforced on-chain) and pays the difference in USDC. **No delivery, no GPU repossession, no seller cooperation.**
- A hedge is only as good as its settlement print: every ACR print ships its **CI** and its **attack-cost-per-bp**, so both sides can read – on-chain – what bending the settle would cost.
- **This loop is live on Arc testnet:** `ACRFutures 0x29d97c62...82642fe` – series 0 (ACR-INF, 10× multiplier, 20% initial margin, socialized-loss clearing) with a **seeded long/short book**, settling against the same ACROracle. Watch it on `/curve`.

THE PRODUCT

Three indices, live on Arc testnet – chain **5042002**

INDEX	MEASURES	LATEST ON-CHAIN VALUE
ACR-INF	Inference – \$/1k tokens	0.49236
ACR-GPU	GPU compute – \$/GPU-sec	0.01110
ACR-DATA	Data egress – \$/MB	0.00209

Every hourly print ships **three numbers**: the rate, a confidence interval, and the **attack-cost-per-bp** (ACR-INF $\approx$ **0.0094 USDC/bp**).

Read live from ACROracle at final submission – posted **hourly, in-cloud, signed by a Circle custody wallet** – and now the print the **futures venue freezes at expiry**.

THE HEADLINE RESULT

\$8,000 wash attack. The attacker got $\leq 2.39\%$.

INDEX	NAIVE VWAP MOVED	ACR MOVED	RESISTANCE
ACR-INF	+110.8%	+0.20%	562x
ACR-GPU	+107.1%	-1.02%	105x
ACR-DATA	+122.5%	+2.39%	51x

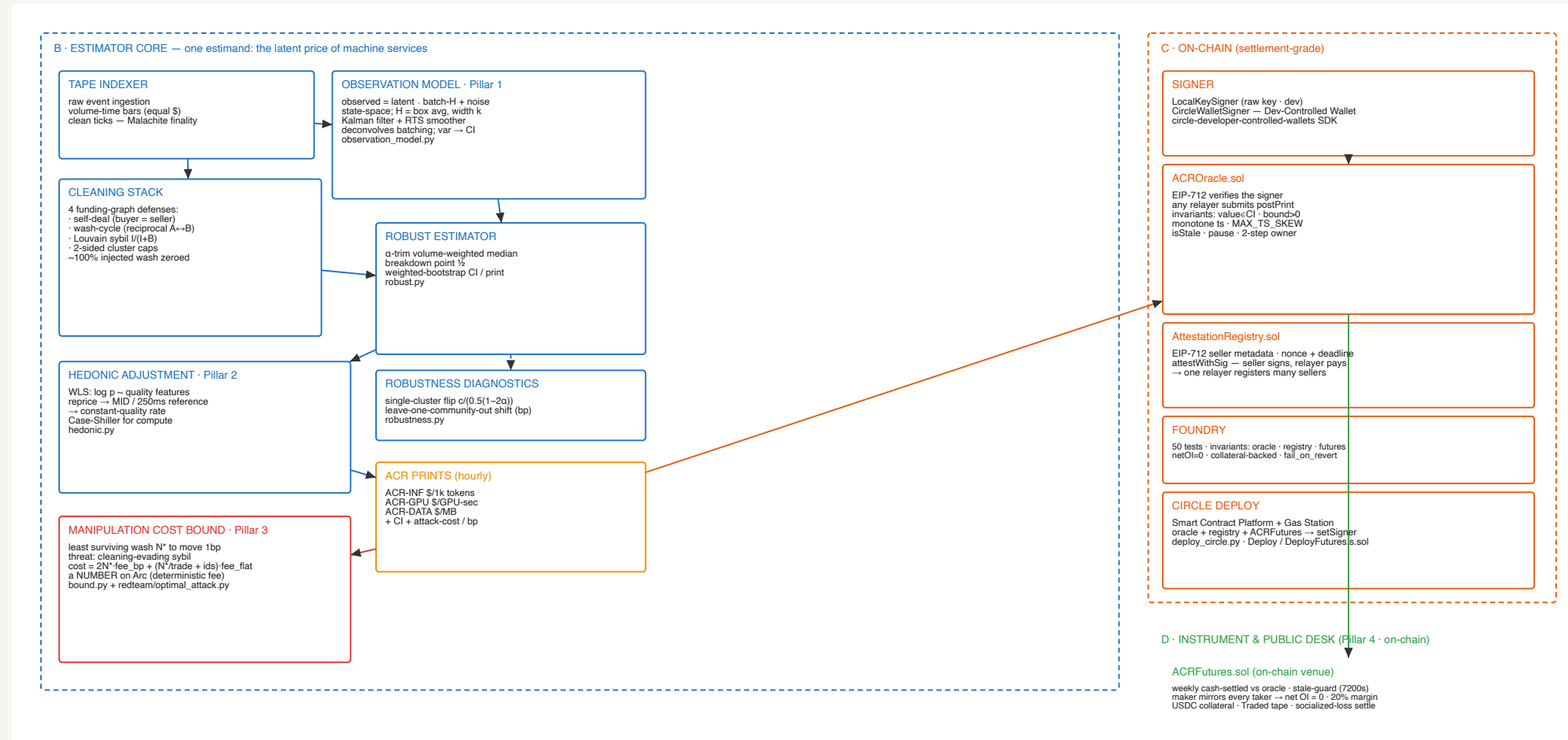
36,000 adversarial authorizations; the attacker burned **\$147.60** in fees. Claims are **CI-gated** (**make eval-gate**) so they cannot silently drift.

Six **mathematical** dependencies – not deployment conveniences

ARC PROPERTY	WHAT IT BUYS THE ESTIMATOR
Malachite deterministic finality	Clean tick timestamps
Gateway: one canonical rail	Complete observation, no selection bias
x402 nanopayments	Agents are the index's paying customers
USDC numeraire	No gas-token deflator in the signal
Deterministic USDC fees	The attack cost is a number, not a distribution
Agent Marketplace	Native machine-to-machine discovery

ARCHITECTURE · ONE PIPELINE

Adversarial exhaust in → settleable print out



Tape → clean (wash · sybil) → deconvolve (Kalman/RTS) → α -trim median + hedonic → print + CI + attack-cost → EIP-712 oracle → futures settle → agents buy the rate back via x402.

Recovering the latent constant-quality price

PILLAR	WHAT IT DOES
Observation model	Kalman + RTS smoother deconvolves Gateway's settlement batching
Cleaning	4 funding-graph defenses (self-deals, wash cycles, sybil clusters, caps) – ~100% of injected wash zeroed
Robust core + hedonic	Trimmed volume-weighted median (breakdown point $\frac{1}{2}$) + bootstrap CI; quality-adjusted via on-chain seller attestations
Manipulation bound	Prices the cheapest surviving attack per bp – validated attainable by <code>redteam/optimal_attack.py</code>

Full spec: `docs/methodology.md` – published before liquidity, the way SOFR was.

Deployed, printing, hedgeable – all seven weeks shipped

CONTRACT / SURFACE	LIVE PROOF
ACROracle 0x4f00e3BD...9092609	Hourly EIP-712 prints, Circle custody signer
AttestationRegistry 0x23ae3E1A...2f63dFb7	4 seller attestations on-chain
ACRFutures 0x29d97c62...82642fe	Series 0 seeded long/short book, cash-settles on the print
Terminal + Seller API	API x402-gated at \$0.0001/query , fail-closed

`make ci` green 2026-07-31: **230 py** · **50 forge** (incl. futures + invariants) · **50 terminal** · **10/10 agent** · resistance **4/4** · interop **12/12** · glossary **332/332** · GitHub CI **4/4**.

Real settlement through **Circle Gateway** – batch-UUID receipts in-repo and on the public `/exchange` tape; Circle's own CLI paid the gate (`payable`).

Provenance labeled on every value – `sim` / `gateway-ref` / `tx` . Reproduce: `make setup && make ci && make demo` .

Attack the index, then hedge on it

1. `/` – the print ticking live: rate + CI + attack cost. New to the jargon? Flip the masthead to **plain** – the whole paper re-sets in plain English.
2. `/attack` – press the button: wash flow floods in, naive VWAP swings **~+110%**, **ACR holds**, and the counter burns the attacker's USDC. Offline twin: `make demo` – one command, ~1 minute.
3. `/curve` – the term structure, the **live futures desk** read from ACRFutures on-chain, and **the Public Desk: trade it yourself**. Open a Circle user-controlled wallet in the browser (your PIN, our gas), take the stake, place a real fill, withdraw it again – the key never leaves your device, so every step is a challenge only your PIN can sign.
4. `/exchange` – real Gateway settlements on the tape; paper trail in `docs/SUBMISSION.md`.

Live: <https://arc-compute-rate.vercel.app> · reader's companion at `/companion`

WHY THIS WINS

The rate machine commerce settles on

- **Judge fit:** ICE – a benchmark administrator – is on Arc's testnet roster; Apollo, BNY, Mastercard are rate-native.
- **The neutrality moat:** Circle can't own the benchmark (the LIBOR lesson) – ACR is a partner, not a feature.
- **The loop closes:** resistant print → on-chain oracle → **cash-settled futures** + x402-paying machine customers → attestations → a better print.
- The first machine-commerce benchmark with a **live derivative venue** – a term structure from day one.

Terminal `arc-compute-rate.vercel.app` · **API** `acr-api-1fto.onrender.com`

Contracts `0x4f00e3BD...` / `0x23ae3E1A...` / `0x29d97c62...` on `testnet.arcscan.app` ·
`docs/SUBMISSION.md`